

Riley Beenders

CONTROLS AUTOMATION | MOTION CONTROL | SOFTWARE-INTEGRATED SYSTEMS

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PROFESSIONAL SUMMARY

Controls and electromechanical R&D leader with 10+ years of multidisciplinary engineering experience, including 6+ years delivering industrial automation, motion-control, and machine-integration systems. Leads architecture, risk analysis, software/hardware integration, validation, troubleshooting, and technical documentation from concept through commercialization.

CORE EXPERTISE

Controls & Motion: Control-system architecture; closed-loop servo and linear-motor systems; Allen-Bradley PLCs; CAN bus; Modbus; I/O and sensor integration; tuning and calibration

Software & Tools: C/C++ (working knowledge); LabVIEW; MATLAB, Simulink, and Simscape; Python; Visual Studio Code (IDE); Git/GitHub

Safety & Delivery: Safety, operational, and software risk analysis; validation workflows; debugging; technical writing; electrical schematics and BOMs; operator training; cross-functional leadership

PROFESSIONAL EXPERIENCE

Proteor / Proteor Printing Solutions | Coplay, PA

January 2025 - Present

Lead, Research & Development Engineer

- Lead the East Coast engineering team and collaborate across Proteor's global product portfolio, owning controls-intensive products from architecture and design through integration, validation, commercialization, and support.
- Led mechanical, electrical, motion-control, and software design of ICARUS-Lite, including dual-voltage 120/240 VAC architecture; documented safety, operational, and software risk analyses and mitigations supporting EU and U.S. requirements.
- Standardized linear-motor architecture across POSEIDON and ICARUS platforms and validated system performance at 4-micron precision, up to 4 m/s, and a 30 kg moving mass.
- Built a LabVIEW/Modbus metrology workflow collecting nine readings every 100 ms, and led machine integration for an operator-facing remote control and diagnostics interface with live monitoring, alerts, and production metrics.

Filament Innovations | Whitehall, PA

May 2020 - January 2025

Chief Technology Officer (Jan 2023 - Jan 2025) | Lead Electro-Mechanical Engineer (Jun 2022 - Jan 2023) | Industrial Engineer (Apr 2021 - Jun 2022) | Engineering Technician (May 2020 - Apr 2021)

- Progressed through four roles to direct product development and technology strategy while remaining hands-on across mechanical, electrical, firmware, software, and manufacturing systems.
- Designed and launched Gen3 POSEIDON with a 1 m³ hybrid extrusion platform, industrial-servo XY motion, braked NEMA 34 Z drives, and a custom gantry.
- Integrated a modular multi-board CAN-bus control architecture and independently testable electronics enclosures, simplifying wiring and debugging while enabling enclosure installation in minutes.
- Led the full lifecycle of large automated systems, including CAD, BOMs, sourcing, assembly, firmware tuning, calibration, operator training, and post-delivery technical support.

ADDITIONAL RELEVANT EXPERIENCE

Blue Mountain Ski Resort - Ski Lift Operator | 2019 - 2021 | Performed motor/control safety checks on passenger lifts and supported maintenance during breakdowns while assisting thousands of guests.

RJB.Engineering - Owner & Engineering Consultant | 2015 - Present | Deliver project-based engineering support for automotive clients and design/manufacture 3D-printing hardware accessories.

Northampton Area School District - Computer/Network Technician | 2016 - 2019 | Automated 2,500+ device deployments with an Arduino-based setup tool and deployed 750+ wireless access points.

EDUCATION & PROFESSIONAL DEVELOPMENT

Penn State University | B.S., Electro-Mechanical Engineering

December 2024

Relevant development: MathWorks MATLAB, Simulink, and Simscape Onramps (2026); Lean Six Sigma Yellow Belt (2025); Penn State Certificate in Engineering Design and Tools (2023)